

Algebraic Coding Theory

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Abstract

Notes taken while studying Algebraic Coding Theory, mostly from Hill's book [1] and Lint's book [2].

Usually while reading books and papers I take handwritten notes in a notebook, this document contains some of them re-written to *LaTeX*.

The proofs may slightly differ from the ones from the books, since I try to extend them for a deeper understanding.

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1 Vector spaces over finite fields, quick recap

Let q be a prime power.

$V(n, q)$: set $GF(q)^n$ of all ordered n -tuples over $GF(q)$. $V(n, q)$ is a vector space.

A set of vectors $\{v_1, v_2, \dots, v_r\}$ is said to be *linear dependent* if $\exists$ scalars $a_1, \dots, a_r$, not all zero, such that

$$a_1v_1 + a_2v_2 + \dots + a_rv_r = 0$$

They are *linear independent* if

$$a_1v_1 + a_2v_2 + \dots + a_rv_r = 0 \xRightarrow{\text{implies}} a_1 = a_2 = \dots = a_r = 0$$

Let C a subspace of $V(n, q)$. Then a subset $\{v_1, v_2, \dots, v_r\} \subseteq C$ is called a *generating set* (or *spanning set*) of C if every vector in C can be expressed as a linear combination of $v_1, \dots, v_r$.

A generating set of C which is also linearly independent is called a *basis* of C .

Theorem 4.3. Suppose $\{v_1, v_2, \dots, v_k\}$ is a basis of a subspace C of $V(n, q)$. Then

- i. every vector of C can be expressed uniquely as a linear combination of the basis vectors.
- ii. C contains exactly q^k vectors.

Proof. i. suppose vector $x \in C$ such that

$$x = a_1v_1 + a_2v_2 + \cdots + a_kv_k$$

$$x = b_1v_1 + b_2v_2 + \cdots + b_kv_k.$$

then

$$(a_1 - b_1)v_1 + (a_2 - b_2)v_2 + \cdots + (a_k - b_k)v_k = 0.$$

But the set $\{v_1, v_2, \dots, v_k\}$ is linearly independent, and so

$$a_i - b_i = 0 \quad \text{for } i = 1, 2, \dots, k,$$

i.e.,

$$a_i = b_i \quad \text{for } i = 1, 2, \dots, k.$$

ii. By (i), the q^k vectors

$$\sum_{i=1}^k a_i v_i \quad (a_i \in \text{GF}(q))$$

are precisely the distinct vectors of C .

□

2 Linear codes

The theorems of this section are from [1] and follow the original enumeration of the book.

Note about notation:

A q -ary linear code $[n, k, d]$ -code is also a q -ary (n, q^k, d) -code, but the opposite is not always true.

2.1 Some definitions

Let q be a prime power. View $(F_q)^n$ as the vector space $V(n, q)$.

Definition (Linear code over $\text{GF}(q)$). A *linear code over $\text{GF}(q)$* is a subspace of $V(n, q)$, with $n \geq 0$.

Thus $C \subseteq V(n, q)$ is a linear code iff

- i. $u + v \in C \quad \forall u, v \in C$
- ii. $au \in C \quad \forall u \in C, a \in \text{GF}(q)$

If C is a k -dimensional subspace of $V(n, q)$, then the linear code C is called an $[n, k]$ -code. (sometimes with minimum distance d : $[n, k, d]$ -code).

Definition (Weight). Weight, $w(x)$, of a vector $x \in V(n, q)$ is the number of non-zero entries of x .

Lemma 5.1. if $x, y \in V(n, q)$, then $d(x, y) = w(x - y)$.

Proof. the vector $x - y$ has non-zero entries in precisely those places where x and y differ. $\square$

Theorem 5.2. let C a linear code, let $w(C)$ be the smallest of weights of the non-zero codewords of C . Then $d(C) = w(C)$.

Proof. $\exists x, y \in C$ s.th. $d(C) = d(x, y)$.

By Lemma 5.1, $d(C) = w(x - y) \geq w(C)$, since $x - y$ is a codeword of the linear code C (ie. $x - y \in C$).

On the other hand, for $x \in C$,
 $w(C) = w(x) = d(x, 0) \geq d(C)$, since $0 \in C$.

Hence, $d(C) \geq w(C)$ and $w(C) \geq d(C) \implies d(C) = w(C)$. $\square$

Definition (Generator matrix (GM)). a $k \times n$ matrix whose rows form a basis of a linear $[n, k]$ -code is called a *generator matrix* (GM) of the code.

2.2 Equivalence

Theorem 5.4. two known $k \times n$ matrices generate equivalent linear $[n, k]$ -codes over $GF(q)$ if one matrix can be obtained from the other by a sequence of operations of the following types:

- R1. permutation of the rows
- R2. multiplication of a row by a non-zero scalar
- R3. addition of a scalar multiple of one row to another
- C1. permutation of the columns
- C2. multiplication of any column by a non-zero scalar

Theorem 5.5. let G a GM of an $[n, k]$ -code. Then by performing the previous operations, G can be transformed to the standard form $[I_k | A]$ where I_k is the $k \times k$ identity matrix, and A is a $k \times (n - k)$ matrix.

2.3 Encoding with a linear code

Let C a $[n, k]$ -code over $GF(q)$ with GM G . C contains q^k codewords $\implies$ can be used to communicate any one of q^k distinct messages.

Identify these messages with the q^k tuples of $V(k, q)$.

Encoding:

msg vector: $u = u_1u_2 \dots u_k$; rows of G : $r_1r_2 \dots r_k$. Encode

$$uG = \sum_{i=1}^k u_i r_i \in C$$

ie. uG is a codeword of C ; a linear combination of the rows of the GM.

Encoding function:

$$\begin{aligned} V(k, q) &\longrightarrow C \subseteq V(n, q) \\ u &\longmapsto uG \end{aligned}$$

If G is in standard form (5.5), $G = [I_k | A]$ with $A = [a_{ij}]$, then encoding u is:

$$x = uG = x_1x_2 \dots x_kx_{k+1} \dots x_n$$

where

$$\begin{aligned} x_i &= u_i \quad \text{for } 1 \leq i \leq k, \text{ message digits} \\ x_{k+i} &= \sum_{j=1}^k a_{ji}u_j \quad \text{for } 1 \leq i \leq n-k, \text{ check digits} \end{aligned}$$

the check digits add redundancy to give protection against noise.

2.4 Decoding with a linear code

Codeword $x = x_1x_2 \dots x_n$, received vector $y = y_1y_2 \dots y_n$, error vector $e = y - x = e_1e_2 \dots e_n$.

2.4.1 Cosets

Cosets here are similar to the ones from typical group theory, just from another perspective.

Definition (Coset). Let C a $[n, k]$ -code over $GF(q)$ and $a \in V(n, q)$. Then the set $a + C = \{a + x | x \in C\}$ is called a *coset* of C .

Definition (Coset leader). the vector having minimum weight in a coset is called the *coset leader*.

Lemma 6.3. Let $a + C$ a coset of C and $b \in a + C$. Then $b + C = a + C$.

Proof. since $b \in a + C \implies b = a + x$ for some $x \in C$. If $b + y \in b + C$ then

$$b + y = (a + x) + y = a + (x + y) \in a + C$$

For the other direction, if $a + x \in a + C$ then

$$a + x = (b - x) + z = b + (z - x) \in b + C$$

hence $a + C \subseteq b + C$.

Therefore $b + C = a + C$. $\square$

Theorem 6.4 (Lagrange). suppose $C \subseteq V(n, q)$ is a $[n, k]$ -code over $GF(q)$. Then

- i. every vector of $V(n, q)$ is in some coset of C
- ii. every coset contains exactly q^k vectors
- iii. two cosets either are disjoint or coincide (partial overlap is impossible)

Proof. i. if $a \in V(n, q)$ then $a = a + 0 \in a + C$

ii. the map

$$\begin{aligned} C &\longrightarrow a + C \quad \forall x \in C \\ x &\longmapsto a + x \end{aligned}$$

is a one-to-one map, hence $|a + C| = |C| = q^k$.

- iii. by contradiction, suppose cosets $a + C$, $b + C$ overlap. Then for some vector v ,

$$v \in (a + C) \cap (b + C)$$

thus for some $x, y \in C$, $v = a + x = b + y$.

Hence $b = a + (x - y) \in a + C$, thus by Lemma 6.3 we have $b + C = a + C$. $\square$

Corollary . 6.4 shows that $V(n, q)$ is partitioned into disjoint cosets of C :

$$V(n, q) = (0 + C) \cup (a_1 + C) \cup \dots \cup (a_s + C)$$

where $s = q^{n-k} - 1$, and by Lemma 6.3 we may take $0, a_1, \dots, a_s$ to be coset leaders.

Example 6.5. Let C be the binary $[4, 2]$ -code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

ie. $C = \{0000, 1011, 0101, 1110\}$.

Then the cosets of C are

$$\begin{aligned} 0000 + C &= C \text{ itself} \\ 1000 + C &= \{1000, 0011, 1101, 0110\} \\ 0100 + C &= \{0100, 1111, 0001, 1010\} = 0001 + C \\ 0010 + C &= \{0010, 1001, 0111, 1100\} \end{aligned}$$

by Lemma 6.3: $0001 \in 0100 + C$, thus $0001 + C = 0100 + C$.

2.5 Probability of error correction (in binary linear codes)

To evaluate how good a code is, must calculate the *probability* that a received vector will be decoded as the codeword which was sent.

Let p be the symbol error probability.

Probability that the error vector is given vector of weight i is $p^i(1-p)^{n-i}$.

Theorem 6.7. Let C a binary $[n, k]$ -code, and for $i = 0, 1, \dots, n$ let α_i denote the number of coset leaders of weight i .

Then the probability $P_{corr}(C)$ that a received vector decoded by means of a standard array is the codeword which was sent is

$$P_{corr}(C) = \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i}$$

Example 6.8. $[n, k]$ -code from 6.5, coset leaders are 0000, 1000, 0100, 0010. Hence $\alpha_0 = 1$, $\alpha_1 = 3$, $\alpha_2 = \alpha_3 = \alpha_4 = 0$, so

$$P_{corr}(C) = (1-p)^4 + 3p(1-p)^3 = (1-p)^3(1+2p)$$

Remark 6.9. if $d(C) = 2t+1$ or $d(C) = 2t+2$, then C can correct any t errors.

Hence every vector of weight $\leq t$ is a coset leader and so $\alpha_i = \binom{n}{i}$ for $0 \leq i \leq t$.

Definition (Rate). $[n, k]$ -code C uses n symbols to send k message symbols; it has *rate* $R(C) = k/n$.

A good code will have a high rate.

Definition (Capacity). the *capacity* $\mathcal{C}(p)$ of a binary symmetric channel with symbol error probability p is

$$\mathcal{C}(p) = 1 + p \cdot \log_2(p) + (1-p) \cdot \log_2(1-p)$$

Theorem 6.12 (Shanon's theorem). channel binary symmetric with symbol error probability p .

Let $R < \mathcal{C}(p)$.

Then for any $\epsilon > 0$, $\exists$ for sufficient large n , an $[n, k]$ -code C of rate $k/n \geq R$ such that $P_{err}(C) < \epsilon$.

Definition (P_{symp}). P_{symp} denotes the average probability that a message symbol is in error after decoding.

Theorem 6.14. let C a binary $[n, k]$ -code and let A_i denote the number of codewords of C of weight i .

Then, if C is used for error detection, the probability of an incorrect message being undetected is

$$P_{\text{undetec}}(C) = \sum_{i=1}^n A_i \cdot p^i (1-p)^{n-i}$$

Example 6.15. for the code of 6.5,

$$P_{\text{undetec}}(C) = \underbrace{1 \cdot p^2 (1-p)^2}_{0101} + \underbrace{2 \cdot p^3 (1-p)^{4-3=1}}_{1011, 1110} = p^2 - p^4$$

2.6 Exercises

Exercise 5.1. is the binary $(11, 24, 5)$ -code linear?

Proof. recall: a q -ary linear $[n, k, d]$ -code is also a q -ary (n, q^k, d) -code, but the opposite is not always true.

$\implies$ $(11, 24, 5)$ -code to be linear it would mean that $24 = q^k$.

Since it is binary, $q = 2$. But $24 \neq 2^k$, so it can not be linear. $\square$

Exercise 5.2. E_n is the code of all even-weight vectors of $V(n, 2)$, which is linear. What are the parameters $[n, k, d]$ of E_n ? Write a GM for E_n in standard form.

Proof.

$$E_n = \{x \in V(n, 2) | w(x) \text{ is even}\}$$

$\longrightarrow$ is the binary even-weight ode (the single parity-check code);
it's linear since

- the zero vector has even weight
- over $GF(2)$, the sum of two even-weight vectors has even weight

Recall: $F_q^n \iff V(n, q)$, so $V(n, 2) \cong F_2^n$

$$\implies E_n = \{(x_1, \dots, x_n) \in F_2^n | x_1 + \dots + x_n = 0\}$$

minimum distance is 2, since

- no word of weight 1 is in E_n
- words of weight 2 are in E_n

Therefore, parameters are $[n, n-1, 2]$
 GM:

$$G = [I_{n-1} | 1]$$

which generates vectors of the form

$$(a_1, \dots, a_{n-1}, a_1 + \dots + a_{n-1})$$

□

Exercise 5.5. Prove that in a *binary linear code*, either all the codewords have even weight or exactly half have even weight and half odd weight.

Proof. let C_e, C_o be subsets of C with even and odd weights respectively.
 So that $C = C_e \cup C_o$ (disjoint union).
 For binary vectors x, y :

$$w(x+y) = w(x) + w(y) \pmod{2}$$

ie. over F_2 , the parity of the weight is additive:

$$even + even = even$$

$$even + odd = odd$$

$$odd + odd = even$$

Now we have two possible cases:

- C has no odd weighted codewords:
 $C_o = \emptyset, \forall x \in C$ x is even weighted
- C has at least one odd-weighted codeword:
 let $u \in C_o$, consider the map

$$\begin{aligned} \phi : C_e &\longrightarrow C_o \\ c &\longmapsto c + u \end{aligned}$$

with $c \in C_e, u \in C_o, \phi(c) \in C_o$.

ϕ is bijective, since $\phi(\phi(c)) = (c + u) + u = c$,

so ϕ is it's own inverse; hence $|C_e| = |C_o|$.

Since $C = C_e \cup C_o$, half codewords are even weighted and half odd.

□

Exercise 6.1. construct standard arrays for codes having each of the GMs:

$$G_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad G_2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Proof. recall: for a binary code with GM G , the code C consists of all binary linear combinations of the rows of G .

For G_1 :

rows: $r_1 = (1, 0)$, $r_2 = (0, 1)$. A codeword is any linear combination:

$$uG_1 = a \cdot r_1 + b \cdot r_2, \text{ with } a, b \in \{0, 1\}$$

4 combinations:

$$0 \cdot r_1 + 0 \cdot r_2 = (0, 0)$$

$$1 \cdot r_1 + 0 \cdot r_2 = (1, 0)$$

$$0 \cdot r_1 + 1 \cdot r_2 = (0, 1)$$

$$1 \cdot r_1 + 1 \cdot r_2 = (1, 0) + (0, 1) = (1, 1)$$

$$\implies C_1 = \{00, 10, 01, 11\}$$

standard array: 00 01 10 11

For G_2 :

rows: $r_1 = (1, 0, 1)$, $r_2 = (0, 1, 1)$. Codeword: $a \cdot r_1 + b \cdot r_2$ with $a, b \in \{0, 1\}$.

4 combinations:

$$0 \cdot r_1 + 0 \cdot r_2 = (0, 0, 0)$$

$$1 \cdot r_1 + 0 \cdot r_2 = (1, 0, 1)$$

$$0 \cdot r_1 + 1 \cdot r_2 = (0, 1, 1)$$

$$1 \cdot r_1 + 1 \cdot r_2 = (1, 0, 1) + (0, 1, 1) = (1, 1, 0)$$

$$\implies C_2 = \{000, 101, 011, 110\}$$

$\implies$ so the 1st row of the standard array is: 000 101 011 110.

Now, pick a non used vector: 001, add it to the code C_2 :

$$001 + C_{2_i}$$

$$001 + 000 = 001$$

$$001 + 101 = 100$$

$$001 + 011 = 010$$

$$001 + 110 = 111$$

$\implies$ 2nd row of the standard array: 001 100 010 111

Thus the standard array is

$$000 \ 101 \ 011 \ 110$$

$$001 \ 100 \ 010 \ 111$$

ie. $2^3 = 8$ vectors, a $V(3, 2)$ vector space.

□

Exercise 6.4. C a binary $[n, k]$ -code with minimum distance $2t + 1$ (or $2t + 2$). Given that p is very small, show that an approximate value of $P_{err}(C)$ is $\left(\binom{n}{t+1} - \alpha_{t+1}\right) p^{t+1}$, where α_{t+1} is the number of coset leaders of C of weight $t + 1$.

Proof. From 6.9, if $d(C) = 2t + 1$ (or $2t + 2$) $\Longrightarrow C$ can correct any t errors.

Hence every vector of weight $\leq t$ is a coset leader and so $\alpha_i = \binom{n}{i}$ for $0 \leq i \leq t$.

$$P_{corr}(C) = \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i} \quad (1)$$

where α_i is the number of coset leaders of weight i

From the $\binom{n}{i}$ vectors of weight i , exactly α_i are coset leaders, so the non coset leaders are $\binom{n}{i} - \alpha_i$. This is when a decoding error happens.

$$\Rightarrow P_{err}(C) = \sum_{i=0}^n \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i}$$

Alternatively from 1:

$$P_{err}(C) = 1 - P_{corr}(C) = 1 - \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i}$$

Notice that

$$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} = (p + (1-p))^n = 1^n = 1$$

thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} - \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i} \\ &= \sum_{i=0}^n \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \end{aligned}$$

Now, from 6.9, $\forall i \leq t$, $\alpha_i = \binom{n}{i}$, thus $\binom{n}{i} - \alpha_i = 0$ for the first t terms in the sum.

Thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^n \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \\ &= \left(\binom{n}{t+1} - \alpha_{t+1} \right) p^{t+1} (1-p)^{n-t+1} + \dots \end{aligned}$$

where the ... represent the terms with p^{t+2} and higher.

Then, since p is very small (by the exercise definition), $(1 - p)^{n-t+1} \equiv 1$, and the other higher powers of p are negligible.

Hence,

$$P_{err}(C) = \left(\binom{n}{t+1} - \alpha_{t+1} \right) \cdot p^{t+1}$$

□

3 Dual code, PCM and Syndrome

3.1 Dual code

Definition (Orthogonal). Inner product:

$$u \cdot v = u_1 v_1 + \dots + u_n \cdot v_n = \sum_{i=1}^n u_i v_i$$

If $u \cdot v = 0$, then u and v are called *orthogonal*.

Lemma 7.1. for any $u, v, w \in V(n, q)$ and $\lambda, \mu \in GF(q)$,

- i. $u \cdot v = v \cdot u$
- ii. $(\lambda u + \mu v) \cdot w = \lambda(u \cdot w) + \mu(v \cdot w)$

Proof. (at exercise 7.1). □

Definition (Dual code). given a linear $[n, k]$ -code C , the *dual code* of C , denoted $C^\perp$, is defined to be the set of those vectors of $V(n, q)$ which are orthogonal to every codeword of C . ie.

$$C^\perp = \{v \in V(n, q) \mid v \cdot u = 0 \forall u \in C\}$$

Lemma 7.2. C a $[n, k]$ -code with GM G . Then $v \in V(n, q)$ belongs to $C^\perp$ iff v is orthogonal to every row of G , ie.

$$v \in C^\perp \iff v \cdot G^T = 0$$

Proof. let rows of G be $r_1, \dots, r_k$, suppose $v \cdot r_i = 0 \forall i$ (ie. v orthogonal to every row of G).

If $u \in C$ (ie. a codeword of C), then $u = \sum_{i=1}^k \lambda_i r_i$ for some scalars λ_i , so that

$$v \cdot u = v \cdot \sum_{i=1}^k \lambda_i r_i = \sum_{i=1}^k \lambda_i \cdot \underbrace{(v \cdot r_i)}_0 = \sum \lambda_i \cdot 0 = 0$$

$\implies$ hence v is orthogonal to every codeword of C , and so is in $C^\perp$. □

Theorem 7.3. C a $[n, k]$ -code over $GF(q)$. Then the dual code $C^\perp$ of C is a linear $[n, n - k]$ -code.

Proof. i. first show that $C^\perp$ is a linear code:

let $v_1, v_2 \in C^\perp$ and $\lambda, \mu \in GF(q)$. Then $\forall u \in C$,

$$(\lambda v_1 + \mu v_2) \cdot u = \lambda(v_1 \cdot u) + \mu(v_2 \cdot u)$$

notice that $v_1, v_2 \in C^\perp$, so

that v_1, v_2 are orthogonal with u

hence $(v_1 \cdot u)$ and $(v_2 \cdot u)$ are 0

$$= \lambda \cdot 0 + \mu \cdot 0 = 0$$

Hence $\lambda \cdot v_1 + \mu \cdot v_2 \in C^\perp$, and so $C^\perp$ is linear, since it is closed under linear combinations.

ii. Next, show that $C^\perp$ has dimension $n - k$:

Let $G = [g_{ij}]$ be a GM of C .

Then by Lemma 7.2, the elements of $C^\perp$ are the vectors $v = (v_1, v_2, \dots, v_n)$, since $v \in C^\perp \iff v \cdot G^T = 0$, satisfying

$$\sum_{j=1}^m g_{ij} v_j = 0, \forall i = [k]$$

$\longrightarrow$ this is a system of k independent homogeneous equations in n unknowns, with solution space $C^\perp$. From linear algebra we know that it has dimension $n - k$. To see it:

if codes C_1 and C_2 are equivalent, then $C_1^\perp, C_2^\perp$ are also equivalent. Hence is enough to show that $\dim(C^\perp) = n - k$, where C is a standard form GM:

$$C = \begin{pmatrix} 1 & \dots & 0 & a_{1,1} & \dots & a_{1,n-k} \\ \vdots & & \vdots & \vdots & & \vdots \\ 0 & \dots & 1 & a_{k,1} & \dots & a_{k,n-k} \end{pmatrix}$$

Then

$$C^\perp = \{(v_1, \dots, v_n) \in V(n, q) \mid v_i + \sum_{j=1}^{n-k} a_{i,j} v_{k+j} = 0, i = 1, 2, \dots, k\}$$

For each of the q^{n-k} choices of $(v_{k+1}, \dots, v_n)$, $\exists! (v_1, v_2, \dots, v_k) \in C^\perp$.

Hence $|C^\perp| = q^{n-k}$, therefore $\dim(C^\perp) = n - k$. □

Theorem 7.5. for any $[n, k]$ -code C , $(C^\perp)^\perp = C$.

Proof. clearly $C \subseteq (C^\perp)^\perp$, since every vector in C is orthogonal to every vector in $C^\perp$.

But $\dim((C^\perp)^\perp) = n - (n - k) = k = \dim(C)$, therefore $C = (C^\perp)^\perp$. □

3.2 PCM (parity-check matrix)

Definition (PCM). A *parity-check matrix* (PCM) H for a $[n, k]$ -code C is a generator matrix of $C^\perp$.

$$\implies H \text{ is a } (n - k) \times n \text{ matrix satisfying } GH^T = 0.$$

From 7.2 and 7.5,
if H a PCM of C , then

$$C = \{x \in V(n, q) \mid xH^T = 0\}$$

$\implies$ Any linear code is completely specified by a PCM.

Intuition . i. PCM: G specifies how to build a codeword, the PCM H specifies the rules a vector must follow to be considered a valid codeword.

ii. dual-code: orthogonal "shadow", see C as a plane through 3D origin; $C^\perp$ is the (normal vector) line emerging from the plane.

Rows of the PCM are the basis for the dual-code $C^\perp \implies$ ie. the rules that define C are themselves a code.

Theorem 7.6. if $G = [I_k | A]$ is the standard form generator matrix of an $[n, k]$ -code C , then a PCM for C is $H = [-A^T \mid I_{n-k}]$.

eg.

$$G = \begin{bmatrix} 1 & \dots & 0 & a_{1,1} & \dots & a_{1,n-k} \\ \vdots & & \vdots & \vdots & & \vdots \\ 0 & \dots & 1 & a_{k,1} & \dots & a_{k,n-k} \end{bmatrix}$$

$$H = \begin{bmatrix} -a_{1,1} & \dots & -a_{1,n-k} & 1 & \dots & 0 \\ \vdots & & \vdots & \vdots & & \vdots \\ -a_{k,1} & \dots & -a_{k,n-k} & 0 & \dots & 1 \end{bmatrix}$$

which, in binary, the minus signs are unnecessary.

Definition (Standard form). a PCM that is $H = [B \mid I_{n-k}]$

3.3 Syndrome

Definition (Syndrome). let H a PCM of a $[n, k]$ -code C .

Then $\forall y \in V(n, q)$, the $1 \times (n - k)$ row vector

$$S(y) = yH^T$$

is called the *syndrome* of y .

Remark . i. for $h_1, h_2, \dots, h_{n-k}$ being rows of H , then

$$S(y) = (y \cdot h_1, y \cdot h_2, \dots, y \cdot h_{n-k})$$

ii. $S(y) = 0 \iff y \in C$

Intuition . Syndrome is the "symptom" of the error. It ignores the valid part of the message and isolates the noise.

$$S(y) = yH^T = (c + e)H^T = \underbrace{cH^T}_0 + eH^T = 0 + eH^T = eH^T$$

Syndrome allows the receiver to identify what went wrong without knowing what was originally sent.

Lemma 7.8. two vectors u, v are the same coset of C iff they have the same syndrome.

Proof. if u, v in the same coset of C

$$\begin{aligned} \implies u + C &= v + C \\ \implies u - v &\in C \\ \implies (u - v)H^T &= 0 \\ \implies uH^T &= vH^T \\ \implies S(u) &= S(v) \end{aligned}$$

□

Corollary 7.9. There is a one-to-one correspondence between cosets and syndromes.

→ for small n , it's fast to locate the received vector y in the array, but for large n : use syndrome to find which coset (ie. which row of the array) contains y .

Example 7.10. (from example 6.5)

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

by 7.6,

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

Hence the syndromes of the coset leaders are

$$S(0000) = 00$$

$$S(1000) = 11$$

$$S(0100) = 01$$

$$S(0010) = 10$$

The standard array becomes:

<i>coset leaders</i>				<i>syndromes</i>
0000	1011	0101	1110	00
1000	0011	1101	0110	11
0100	1111	0001	1010	01
0010	1001	0111	1100	10

so, decoding algorithm: when a vector y is received, calculate $S(y) = yH^T$, locate $S(y)$ in the *syndromes* column of the array.

Locate y in the corresponding row and decode as the codeword at the top of the column containing y .

→ in practice: *syndrome lookup table*:

syndrome z	coset leader $f(z)$
00	0000
11	1000
01	0100
10	0010

decoding procedure:

1. receive y , calculate $S(y) = yH^T$
2. let $z = S(y)$, locate z in the first column of the lookup table
3. decode y as $y - f(z)$

3.4 Incomplete decoding

For $d(C) = 2t + 1$ (or $2t + 2$), guarantee the correction of $\leq t$ errors in any codeword and also detect some cases of more than t errors.

→ Arrange the cosets in order of increasing weight of the coset leaders.

Divide the array into a top part and a bottom part; top: cosets whose leaders have weights $\leq t$.

- If y in top part: decode as usual (assuming $\leq t$ errors).
- If y in bottom part: conclude that $> t$ errors have occurred, ask for retransmission.

3.5 Exercises

Exercise 7.1. prove Lemma 7.1

Proof. For any $u, v, w \in V(n, q)$ and $\lambda, \mu \in GF(q)$,

- i. $u \cdot v = v \cdot u$

$$u \cdot v = \sum_{i=1}^n u_i \cdot v_i = \sum_{i=1}^n v_i \cdot u_i = v \cdot u$$

ii. $(\lambda u + \mu v) \cdot w = \lambda(u \cdot w) + \mu(v \cdot w)$

$$\begin{aligned}
(\lambda u + \mu v)w &= \sum (\lambda u_i + \mu v_i)w_i \\
&= \sum (\lambda u_i w_i + \mu v_i w_i) \\
&= \sum (\lambda u_i w_i) + \sum (\mu v_i w_i) \\
&= \lambda \sum (u_i w_i) + \mu \sum (v_i w_i) \\
&= \lambda \cdot (u \cdot w) + \mu \cdot (v \cdot w)
\end{aligned}$$

□

Exercise 7.2. prove that if E_n is the binary even weight code of length n , then $E_n^\perp$ is the repetition code of length n .

Proof. from exercise 5.2:

$$E_n = \{x \in V(n, 2) \mid w(x) \text{ is even}\} = \{(x_1, \dots, x_n) \in F_2^n \mid x_1 + \dots + x_n = 0\}$$

GM of E_n :

$$G = \left[I_{n-1} \mid \begin{array}{c} 1 \\ \vdots \\ 1 \end{array} \right]$$

thus, by 7.6, PCM is $H = [1 \ 1 \ \dots \ 1]$.

If H is a PCM for C , then the rows of H generate $C^\perp$.

Therefore

$$E_n^\perp = \langle \underbrace{[1 \ 1 \ \dots \ 1]}_{\text{generator}} \rangle$$

then it's codewords are:

$$0 \cdot (1, \dots, 1) = (0, \dots, 0)$$

$$1 \cdot (1, \dots, 1) = (1, \dots, 1)$$

$$\implies E_n^\perp = \{00 \dots 00, 11 \dots 1\}$$

ie. the repetition code of length n .

□

Exercise 7.4. for a binary linear code with PCM H , show that the transpose of the syndrome of a received vector is equal to the sums of those columns of H corresponding to where the errors occurred.

Proof. Let e the error, y the vector received, and x the original codeword. Then $y = x + e$.

Then

$$S(y) = yH^T = (x + e)H^T = \underbrace{xH^T}_{\substack{0, \text{ since } H \\ \text{is PCM of } C \\ \text{and } x \in C}} + eH^T = eH^T$$

So,

$$S(y)^T = e^T H = \sum_{j=1}^n e_j \cdot H_j$$

where H_j is the j -th column of H . □

Exercise 7.6. Let C be the ternary linear code with GM

$$G = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{bmatrix}$$

- (a) Find a generator matrix in standard form.
- (b) Find a parity check matrix (PCM) for C in standard form.
- (c) Use syndrome decoding to decode the received vectors: 2122, 1201, 2222

Proof. (a)

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

thus $G = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$ is PCM in standard form.

- (b) since $G = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$ then

$$H = \begin{bmatrix} -2 & -2 & 1 & 0 \\ -2 & -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

- (c)

$$S(y) = yH^T = S(y) = (y \cdot h_1, y \cdot h_2).$$

w/ h_i the rows of H .

compute the lookup table (syndrome table):

e	$S(e)$
0000	00
1000	11
2000	22
0100	12
0200	21
0010	10
0020	20
0001	01
0002	02

left: weight-1 errors, since we're over F_3 ; right: coset leaders for the 9 cosets.

i. $y = 2121$:

$$S(y) = \begin{bmatrix} 2 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = [5, 5] \equiv [2, 2].$$

$S(y) = [2, 2]$, corresponds (in the table) to $e = 2000$.

$$\implies c = y - e = 2121 - 2000 = 0121$$

ii. $y = 1201$

$$S(y) = \begin{bmatrix} 1 & 2 & 0 & 1 \end{bmatrix} H^T = [3, 6] = [0, 0].$$

corresponds to $e = 0000$

$$\implies c = y - e = 1201$$

iii. $y = 2222$

$$S(y) = \begin{bmatrix} 2 & 2 & 2 & 2 \end{bmatrix} H^T = [6, 8] = [0, 2].$$

corresponds to $e = 0002$

$$\implies c = y - e = 2222 - 0002 = 2220$$

□

4 Hamming Codes

Definition (Binary Hamming Code). let r a positive integer, H a $r \times (2^r - 1)$ matrix whose columns are the distinct non-zero vectors of $V(r, 2)$.

The code having H as its PCM is called a *binary Hamming code*, denoted $Ham(r, 2)$.

Remark . $Ham(r, 2)$ has length $n = 2^r - 1$ and dimension $k = n - r$.

Redundancy of the code: $r = n - k$, the number of check symbols in each codeword.

Exercise 8.1. i. $r = 2$: $H = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

by Theorem 7.6, $G = \begin{bmatrix} 1 & 1 & \underbrace{1}_I \end{bmatrix}$

so $Ham(2, 2)$ is just the binary triple repetition code.

ii. $r = 3$: a PCM for $Ham(3, 2)$ is

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

(have taken columns in the natural order of increasing binary numbers from 1 to 7)

H in standard form: rearrange columns:

$$H = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

by Theorem 7.6, a generator matrix for $Ham(3, 2)$ is

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Theorem 8.2. the binary Hamming code $Ham(r, 2)$, for $r \geq 2$,

1. is a $[2^r - 1, 2^r - 1 - r]$ -code
2. has minimum distance 3 (hence is single-error-correcting)
3. is a *perfect code*

Lemma (Decoding with a binary Hamming code). $Ham(r, 2)$ is a perfect single-error-correcting code $\implies$ the coset leader are precisely the 2^r ($= n + 1$) vectors of $V(n, 2)$ of weight ≤ 1 .

Let j -th column of H be the binary representation of j . Decoding algorithm:

1. receive vector y , calculate its syndrome: $S(y) = yH^T$
2. if $S(y) = 0$, assume $y = c$ (original sent codeword)
3. if $S(y) \neq 0$, assuming a single error, $S(y)$ gives the binary representation of the error position, and the error can be corrected.

Example . $H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$, if $y = 1101011$ then

$$S(y) = y \cdot H^T = 110$$

$\implies$ 6-th position, and so we decode y as 1101001

4.1 Extended binary Hamming codes

$\hat{Ham}(r, 2)$, obtained from $Ham(r, 2)$ by adding an overall parity-check.

Not ideal for complete decoding, but good for incomplete decoding (correct any single error and detect any double error).

Let H be a PCM for $Ham(r, 2)$. A PCM $\hat{H}$ for $\hat{Ham}(r, 2)$:

$$\hat{H} = \begin{bmatrix} & & & 0 \\ & H & & \vdots \\ & & & 0 \\ 1 & \cdots & 1 & 1 \end{bmatrix}$$

If H is taken with columns in increasing order of binary numbers, we have a neat incomplete decoding algorithm:

Example 8.3. $r = 3$, $\hat{Ham}(3, 2)$. Received y , $S(y) = y\hat{H}^T = (s_1, s_2, s_3, s_4)$

- i. if $s_4 = 0$ and $(s_1, s_2, s_3) = 0$: assume no errors
- ii. if $s_4 = 0$ and $(s_1, s_2, s_3) \neq 0$: assume ≥ 2 errors and ask retransmission
- iii. if $s_4 = 1$ and $(s_1, s_2, s_3) = 0$: assume 1 error in the last position
- iv. if $s_4 = 1$ and $(s_1, s_2, s_3) \neq 0$: assume 1 error in the j -th position, where j is the number binary-represented by (s_1, s_2, s_3)

Theorem 8.4. suppose C a linear $[n, k]$ -code over $GF(q)$ with PCM H . Minimum distance of C is d iff any $d - 1$ columns of H are linearly independent but some d columns are linearly dependent.

Proof. by 5.2, min. distance of C (ie. $d(C)$) = smallest of the weights of the non-zero codewords.

Let $x = x_1x_2 \dots x_n \in V(n, q)$, then

$$\begin{aligned} x \in C &\iff xH^T = 0 \\ &\iff x_1H_1 + x_2H_2 + \dots + x_nH_n = 0 \end{aligned}$$

where H_i denote the columns of H .

$\implies$ each codeword x of weight d corresponds with a set of d linearly dependent columns of H .

On the other hand, if $\exists$ a set of $d - 1$ lin. dependent columns of H , say $H_{i_1}, H_{i_2}, \dots, H_{i_{d-1}}$, then $\exists x_{i_1}, \dots, x_{i_{d-1}}$ scalars, not all zero, such that

$$x_{i_1}H_{i_1} + x_{i_2}H_{i_2} + \dots + x_{i_{d-1}}H_{i_{d-1}} = 0$$

But then, the vector

$$x = (0 \dots 0x_{i_1}0 \dots x_{i_2}0 \dots 0x_{i_{d-1}}0 \dots 0)$$

having x_{i_j} in the j -th position for $j \in [d-1]$, would satisfy $xH^T = 0$, and so would be a non-zero codeword of weight $< d$. $\square$

4.2 q-ary Hamming codes

For C a linear code with $d(C) = 3$ (min.dist), any 2 columns of a PCM H must be linearly independent

$\implies$ columns of H must be non-zero, and no column can be a scalar multiple of another

Any $0 \neq v \in V(r, q)$ vector has $q-1$ non-zero scalar multiples, forming the set

$$\{\lambda v \mid \lambda \in GF(q), \lambda \neq 0\}$$

$\longrightarrow$ the q^r-1 non-zero vectors of $V(r, q)$ may be partitioned into $\frac{q^r-1}{q-1}$ such sets, called *classes*.

$\longrightarrow$ two vectors are scalar multiples of each other if they are in the same class

By choosing one vector from each class, obtain a set of $\frac{q^r-1}{q-1}$ vectors, where not two are linearly dependent.

By Theorem 8.4, taking these as columns of h gives a PCM for a

$$\left[\frac{q^r-1}{q-1}, \frac{q^r-1}{(q-1)-r}, 3 \right] \text{-code}$$

$\implies$ called *q-ary Hamming code*, $Ham(r, q)$.

$\longrightarrow$ PCM is unique up to permutation of columns

$\implies$ Hamming codes are linear codes

Theorem 8.6. $Ham(r, q)$ is a perfect single-error-correcting code.

Proof. $Ham(r, q)$ is constructed to be a $(n, M, 3)$ -code, with $n = \frac{q^r-1}{q-1}$ and $M = q^{n-r}$.

With $t = 1$, the LHS of the sphere-packing bound becomes

$$q^{n-r} \cdot (1 + n(q-1)) = q^{n-r} \cdot (1 + q^r - 1) = q^n$$

which is the RHS, so $Ham(r, q)$ is a perfect code. $\square$

Decoding q-ary Hamming code Since Ham is perfect single-error correcting code, the coset leaders, other than 0, are the vectors of weight 1.

Syndrome of such coset leader:

$$S(0 \dots 0 b 0 \dots 0) = (0 \dots 0 b 0 \dots 0) \cdot H^T = b \cdot H_j^T$$

where H_j is the j -th column of H , and b is at the j -th position of y .

Decode:

- given received y , calculate $S(y) = yH^T$
- if $S(y) = 0$, assume no errors
- if $S(y) \neq 0$, then $S(y) = b \cdot H_j^T$ for some b and j
correct by: subtract b from the j -th entry of y .

Example:

$$q = 5, H = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 3 & 4 \end{bmatrix}, y = 203031$$

Then,

$$\begin{aligned} S(y) &= y \cdot H^T = 2 \cdot H_1^T + 3 \cdot H_3^T + 3 \cdot H_5^T + 1 \cdot H_6^T \\ &= 2 \cdot [0 \ 1] + 3 \cdot [1 \ 1] + 3 \cdot [1 \ 3] + 1 \cdot [1 \ 4] \\ &= [0 \ 2] + [3 \ 3] + [3 \ 9] + [1 \ 4] \\ &= [7 \ 8] = [2 \ 3] = \underbrace{2}_b \cdot \underbrace{[1 \ 4]}_{H_6^T} \pmod{5} \end{aligned}$$

$\implies$ error is in position 6-th, with magnitude $b = 2$
 $\implies$ subtract b from the j -th entry of y : $1 - 2 \pmod{5} = -1 \pmod{5} = 4$,
therefore $y' = 203034$.

5 MLCT - Main Linear Coding Theory problem

A good (n, M, d) -code has

- small n , for fast transmission of messages
- large M , to enable transmission of a wide variety of messages
- large d , to correct many errors

Denote by $A_q(n, d)$ the largest M such that $\exists$ a q -ary (n, M, d) -code.

Theorem 2.7. Suppose d is odd. Then a binary (n, M, d) -code exists iff a binary $(n + 1, M, d + 1)$ -code exists.

Proof. □

Corollary 2.8. If d is odd, then $A_2(n + 1, d + 1) = A_2(n, d)$.

Equivalently, if d is even, then $A_2(n, d) = A_2(n - 1, d - 1)$.

Definition (Sphere). $\forall u \in F_q^n$ (vector) and any integer $r \geq 0$, the *sphere* of radius r and centre u , denoted $S(u, r)$, is the set

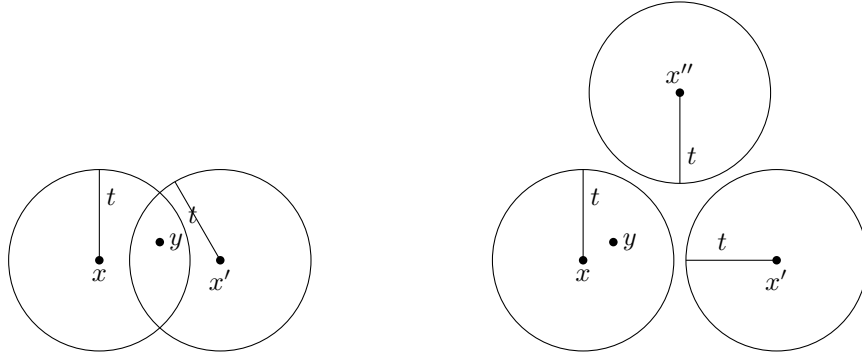
$$\{v \in F_q^n \mid d(u, v) \leq r\}$$

Theorem 1.9 & 2.12. a code C can

- i. detect up to s errors in any codeword if $d(C) \geq s + 1$
- ii. correct up to t errors in any codeword if $d(C) \geq 2t + 1$

This is, the spheres of radius t centered on the codewords of C are disjoint (ie. have no overlap).

So, if $\leq t$ errors occur, then the received vector y may be different from the sent codeword x , hence being different from the centre of the sphere $S(x, t)$, but can not 'escape' from the sphere, so is 'drawn back' to x by nearest neighbour decoding.



Definition (Perfect codes). A code which achieves the sphere-packing bound is called a *perfect code*. Thus, for a perfect t -error-correcting code, the M spheres of radius t centred on codewords 'fill' the whole space F_q^n without overlapping.

In other words, every vector in F_q^n is at distance $\leq t$ from exactly one codeword.

Theorem (MLCT problem (Version 1)). For given length n , find the maximum dimension k such that $\exists$ an $[n, k, d]$ -code over $GF(q)$. (Then, for this k , $B_q(n, d) = q^k$).

Remark . *redundancy* r of an $[n, k, d]$ -code is just $r = n - k$ (the number of check symbols in a codeword).

Theorem (MLCT problem (Version 2)). For given redundancy r , find the maximum length n such that $\exists$ an $[n, n - r, d]$ -code over $GF(q)$.

Definition $((n, s)$ -set). a (n, s) -set in $V(r, q)$ is a set of n vectors in $V(r, q)$ with the property that any s of them are linearly independent.

Denote $max_s(r, q)$ the largest value of n for which $\exists$ a (n, s) -set in $V(r, q)$. ie.

- (n, s) -set: n vectors with s of them lin. independent

- $max_s(r, q)$: max n for which $\exists (n, s)$ -set

Definition (Packing problem). The *packing problem* for $V(r, q)$ is that of determining the values of $max_s(r, q)$ and the optimal (n, s) -sets.

Theorem 14.1. There exists an $[n, n - r, d]$ -code over $GF(q)$ iff $\exists$ an $(n, d - 1)$ -set in $V(r, q)$.

Proof. let C a $[n, n - r, d]$ -code over $GF(q)$ with PCM H . Then by Theorem 8.4, the columns of H form a $(n, d - 1)$ -set in $V(r, q)$.

On the other hand, let K a $(n, d - 1)$ -set in $V(r, q)$. If we form a $r \times n$ matrix H with the vectors of K as its columns, by Theorem 8.4, H is the PCM of a $[n, n - r]$ -code whose minimum distance is at least d . $\square$

Corollary 14.2. for given values q, d, r , the largest value of n for which there exists a $[n, n - r, d]$ -code over $GF(q)$ is $max_{d-1}(r, q)$.

$\implies$ the MLCT problem (V2) is the same as the packing problem of finding $max_{d-1}(r, q)$.

5.1 The projective geometry $PG(r-1, q)$

Let the vector space

$$V(r, q) = \{(a_1, a_2, \dots, a_r) \mid a_i \in GF(q)\}$$

Definition (Projective Geometry). $PG(r - 1, q)$

- *points* are the one-dimensional subspaces of $V(r, q)$
- *lines* are the two-dimensional subspaces of $V(r, q)$

Each point $P \in PG(r - 1, q)$, as a subspace of $V(r, q)$ of dimension 1, is generated by a single non-zero vector.

So if $a = (a_1, a_2, \dots, a_r) \in P$, then $P = \{\lambda a \mid \lambda \in GF(q)\}$.

$\implies$ for $a = (a_1, a_2, \dots, a_r)$, $b = (b_1, b_2, \dots, b_r) \in P$, then

$$a = b \text{ in } PG(r - 1, q) \text{ iff } a = \lambda b \text{ in } V(r, q)$$

for some non-zero scalar λ .

Lemma 14.6. in $PG(r - 1, q)$,

1. number of points is $\frac{q^r - 1}{q - 1}$
2. any two points lie on exactly one line
3. each line contains exactly $q + 1$ points
4. each point lies on $\frac{q^{r-1} - 1}{q - 1}$ lines

Proof. 1. number of points is $\frac{q^r-1}{q-1}$:

since each of the $q^r - 1$ nonzero vectors in $V(r, q)$ has $q - 1$ nonzero scalar multiples, the number of points of $PG(r - 1, q)$ is $\frac{q^r-1}{q-1}$

2. any two points lie on exactly one line:

if a, b distinct points of $PG(r - 1, q)$, then the unique line through them consists of the points $\lambda a + \mu b$, where λ, μ are scalars not both zero.

3. each line contains exactly $q + 1$ points:

in (ii) there are $q^2 - 1$ choices for the pair (λ, μ) , but we're identifying scalar multiples, so the number of distinct points on the line is

$$\frac{q^2 - 1}{q - 1} = q + 1$$

4. each point lies on $\frac{q^{r-1}-1}{q-1}$ lines:

let t be the number of lines on which a point P lies. Let the set

$$X = \{(Q, L) \mid Q \text{ is a point } \neq P, L \text{ is a line containing both } P \text{ and } Q\}$$

Count the members of X in two ways:

for each of the $\frac{q^r-1}{q-1} - 1$ choices of Q , $\exists!$ line containing P and Q . Thus,

$$|X| = \frac{q^r - 1}{q - 1} - 1 = \frac{q^r - q}{q - 1}$$

On the other hand, for each of the t lines through P , there are (by (iii)) q points Q other than P lying on L .

Thus, $|X| = tq$,

$$\implies tq = \frac{q^r - q}{q - 1}, \quad \implies t = \frac{q^{r-1} - 1}{q - 1}$$

□

Definition (Projective plane). The $PG(2, q)$ is called the *projective plane* over $GF(q)$.

6 MDS Codes

Theorem 15.1. a $[n, n - r, d]$ -code satisfies $d \leq r + 1$.

Proof. let C a $[n, n - r, d]$ -code, let $G = [I_{n-r} \mid A]$ be a standard form GM of C .

Since A has r columns, the codewords which are rows of G have weight $\leq r + 1$, ie. $w(C) \leq r + 1$.

The result follows from Theorem 5.2,

$$\begin{aligned} \implies w(C) &= d(C) \\ \implies d(C) &\leq r + 1. \end{aligned}$$

□

Definition (MDS). a $[n, n - r, r + 1]$ -code,

ie. a linear code of redundancy r whose min. distance is $r + 1$,
is called a *maximum distance separable code*, MDS.

→ by Theorem 14.1, max length of a MDS code over $GF(q)$ is $\max_r(r, q)$,
the largest size of a (n, r) -set in $V(r, q)$.

Conjecture 15.2. if $2 \leq r \leq q$, then $\max_r(r, q) = q + 1$
(except $\max_3(3, q) = \max_{q-1}(q - 1, q) = q + 2$ if $q = 2^h$)

Theorem 15.3. if $r \geq q$, then $\max_r(r, q) = r + 1$. Any MDS code of redundancy
 $r \geq q$ is equivalent to a repetition code of length $r + 1$.

Proof. repetition code of length $r + 1$: a $[r + 1, 1, r + 1]$ -code with GM $[1 \ 1 \ \dots \ 1]$.

Hence $\max_r(r, q) \geq r + 1$.

Also, any $[r + 1, 1, r + 1]$ -code is equivalent to a repetition code.

Suppose $r \geq q$ and by contradiction that $\max_r(r, q) \geq r + 2$.

Then, $\exists$ a $[r + 2, 2, r + 1]$ -code C over $GF(q)$.

→ C must be equivalent to a code having GM:

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & \dots & 1 \\ 0 & 1 & a_1 & a_2 & \dots & a_r \end{bmatrix}$$

in order that any linear combination of the rows of G has weight at least
 $r + 1$ with $0 \neq a_i \in GF(q)$.

This implies $r \leq q - 1$, contradiction.

□

Remark . the only binary MDS codes are

$V(n, 2)$, the even weighted codes E_n , and repetition codes.

Theorem 15.4. let $2 \leq r \leq q$, let $a_1, a_2, \dots, a_{q-1} \in GF(q)$ nonzero.

Then,

$$H = \begin{bmatrix} 1 & 1 & \dots & 1 & 1 & 0 \\ a_1 & a_2 & \dots & a_{q-1} & 1 & 0 \\ a_1^2 & a_2^2 & \dots & a_{q-1}^2 & 1 & 0 \\ \vdots & \vdots & & \vdots & \vdots & \vdots \\ a_1^{r-1} & a_2^{r-1} & \dots & a_{q-1}^{r-1} & 0 & 1 \end{bmatrix}$$

is the PCM of a MDS $[q + 1, q + 1 - r, r + 1]$ -code.

Equivalently, the columns of H form a $(q + 1)$ -arc in $PG(r - 1, q)$.

Proof. the determinant of a matrix formed by any r columns of H is equal to the determinant of a Vandermonde matrix, and so is non-zero (see a proof of this at section 6.1). Thus any r columns of H are linearly independent. $\square$

Corollary 15.7. The dual code of a MDS code is also MDS.

Corollary 15.8. $\exists$ a MDS $[n, k]$ -code over $GF(q)$ iff $\exists$ a MDS $[n, n - k]$ -code over $GF(q)$.

6.1 Vandermonde determinant

This section contains a simple proof of the Vandermonde matrix determinant. The proof was found in [3], here it is written with small modifications and expanded. A different proof can be found in [4].

Lemma (Vandermonde determinant).

$$\det V(x_1, \dots, x_n) = \det \begin{pmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \dots & x_2^{n-1} \\ 1 & x_3 & x_3^2 & \dots & x_3^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^{n-1} \end{pmatrix} = \prod_{j < i} (x_i - x_j)$$

Proof. The equation is true for $n = 2$, as we have:

$$\det V(x_1, x_2) = \det \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \end{pmatrix} = x_2 - x_1$$

We now assume that $n \geq 3$. Starting with column n and ending at column 2, we successively apply the following operation: subtract x_1 times column $i - 1$ from column i . The determinant remains the same and the resulting matrix is:

$$\begin{aligned} &= \det \begin{pmatrix} 1 & x_1 - (x_1 \cdot 1) & x_1^2 - (x_1 \cdot x_1) & \dots & x_1^{n-1} - (x_1 \cdot x_1^{n-2}) \\ 1 & x_2 - (x_1 \cdot 1) & x_2^2 - (x_1 \cdot x_2) & \dots & x_2^{n-1} - (x_1 \cdot x_2^{n-2}) \\ 1 & x_3 - (x_1 \cdot 1) & x_3^2 - (x_1 \cdot x_3) & \dots & x_3^{n-1} - (x_1 \cdot x_3^{n-2}) \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n - (x_1 \cdot 1) & x_n^2 - (x_1 \cdot x_n) & \dots & x_n^{n-1} - (x_1 \cdot x_n^{n-2}) \end{pmatrix} \\ &= \det \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & (x_2 - x_1) & x_2(x_2 - x_1) & \dots & x_2^{n-2}(x_2 - x_1) \\ 1 & (x_3 - x_1) & x_3(x_3 - x_1) & \dots & x_3^{n-2}(x_3 - x_1) \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & (x_n - x_1) & x_n(x_n - x_1) & \dots & x_n^{n-2}(x_n - x_1) \end{pmatrix} \end{aligned}$$

Applying the Laplace expansion along the first row and factoring out the scalar multiples $(x_i - x_1)$ in each column, we obtain:

$$= \det \begin{pmatrix} x_2 - x_1 & x_2^2 - x_1 x_2 & x_2^3 - x_1 x_2^2 & \dots & x_2^{n-1} - x_1 x_2^{n-2} \\ x_3 - x_1 & x_3^2 - x_1 x_3 & x_3^3 - x_1 x_3^2 & \dots & x_3^{n-1} - x_1 x_3^{n-2} \\ \vdots & \vdots & \vdots & & \vdots \\ x_n - x_1 & x_n^2 - x_1 x_n & x_n^3 - x_1 x_n^2 & \dots & x_n^{n-1} - x_1 x_n^{n-2} \end{pmatrix}$$

in each row, factor out $(x_i - x_1)$

$$= \det \begin{pmatrix} 1(x_2 - x_1) & x_2(x_2 - x_1) & x_2^2(x_2 - x_1) & \dots & x_2^{n-2}(x_2 - x_1) \\ 1(x_3 - x_1) & x_3(x_3 - x_1) & x_3^2(x_3 - x_1) & \dots & x_3^{n-2}(x_3 - x_1) \\ \vdots & \vdots & \vdots & & \vdots \\ 1(x_n - x_1) & x_n(x_n - x_1) & x_n^2(x_n - x_1) & \dots & x_n^{n-2}(x_n - x_1) \end{pmatrix}$$

$$= \prod_{2 \leq i \leq n} (x_i - x_1) \det \begin{pmatrix} 1 & x_2 & x_2^2 & \dots & x_2^{n-2} \\ 1 & x_3 & x_3^2 & \dots & x_3^{n-2} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^{n-2} \end{pmatrix}$$

now, by induction

$$= \prod_{i=2}^n (x_i - x_1) \cdot \prod_{2 \leq j < i \leq n} (x_i - x_j) = \prod_{1 \leq j < i \leq n} (x_i - x_j)$$

□

7 Cyclic Codes

This section contains notes both from Hill [1] and Lint [2] books. Hill's notes are enumerated as $X.Y$ while Lint's notes as $X.Y.Z$.

7.1 Recap on ideals

More about ideals at the *Commutative Algebra notes* [5].

In the cyclic codes context, we're interested in ideals because

- cyclic codes are ideals of $R_n = \mathbb{F}_q[x]/(x^n - 1)$.
- R_n is a PID: every ideal in R_n is principal.

Definition (ideal). $I \subset R$ (R ring) such that $0 \in I$ and $\forall x \in I, r \in R, xr, rx \in I$.

ie. I absorbs products in R .

Definition (prime ideal). if $a, b \in R$ with $ab \in P$ and $P \neq R$ (P a prime ideal), implies $a \in P$ or $b \in P$.

Definition (principal ideal). generated by a single element, (a) .

(a) : principal ideal, the set of all multiples xa with $x \in R$.

Definition (prime spectrum - $\text{Spec}(A)$). set of prime ideals of A . ie.

$$\text{Spec}(A) = \{P \mid P \subset A \text{ is a prime ideal}\}$$

Definition (integral domain). Ring in which the product of any two nonzero elements is nonzero.

ie. no zerodivisors.

ie. $\forall 0 \neq a, 0 \neq b \in A, ab \neq 0 \in A$.

Every field is an integral domain, not the converse.

Definition (principal ideal domain - PID). integral domain in which every ideal is principal. ie. ie. $\forall I \subset R, \exists a \in I$ such that $I = (a) = \{ra \mid r \in R\}$.

7.2 Cyclic Codes intro

Definition (cyclic code). a code C is cyclic if

- i. C is a linear code
- ii. any cyclic shift of a codeword is also a codeword
ie. when $a_0a_1 \dots a_{n-1} \in C$,
then so $a_{n-1}a_0a_1 \dots a_{n-2} \in C$.

Remark (Notation). From now on,

$$F_q = \mathbb{F}_q = GF(q)$$

$F_q[x]$ is a ring; and for $f(x)$ irreducible in $F_q[x]$, then $\frac{F_q[x]}{f(x)}$ is a field.

From now on,

$$f(x) = x^n - 1 \implies F_q[x]/(x^n - 1) = R_n$$

Observe:

since $x^n \cong 1 \pmod{x^n - 1}$, to reduce any polynomial by modulo $x^n - 1$ can just replace x^n by 1, x^{n+1} by x , x^{n+2} by x^2 , etc.

Correspondence:

$$a_0a_1 \dots a_{n-1} \in V(n, q) \iff a(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} \in R_n$$

In R_n ,

$$\begin{aligned} x \cdot a(x) &= a_0 \cdot x + a_1x^2 + \dots + \underbrace{a_{n-1}x^n}_{=a_{n-1} \cdot 1} \\ &= a_{n-1} + a_0x + \dots + a_{n-2} \cdot x^{n-1} \in R_n \end{aligned}$$

ie. $a_{n-1}a_0a_1 \dots a_{n-2} \in V(n, q)$

In R_n :

→ multiply by x corresponds to performing a single cyclic shift

→ multiply by x^m corresponds to a cyclic shift through m positions

Theorem 12.6 (Algebraic characterization of cyclic codes). a code $C \in R_n$ is a cyclic code iff

- i. $a(x), b(x) \in C \implies a(x) + b(x) \in C$
- ii. $a(x) \in C$ and $r(x) \in R_n \implies r(x)a(x) \in C$
ie. C closed under multiplication by any element of R_n

Proof. i. let $C \in R_n$ cyclic code, then C is linear, so (i) holds.

- ii. let $a(x) \in C$ and $r(x) = r_0 + r_1x + \dots + r_{n-1}x^{n-1} \in R_n$.

Since mult. by x corresponds to a cyclic shift, then $x \cdot a(x) \in C$, thus

$$x \cdot (x \cdot a(x)) = x^2 \cdot a(x) \in C$$

and so on.

Hence,

$$r(x) \cdot a(x) = r_0 \cdot a(x) + r_1 \cdot x \cdot a(x) + \dots + r_{n-1} \cdot x^{n-1} \cdot a(x) \in C$$

since each summand is in C ; thus (ii) holds.

Conversely, suppose (i) and (ii) hold;

taking $r(x)$ to be a scalar, the conditions imply that C is linear.

Taking $r(x) = x$, in (ii) shows that C is cyclic.

□

Remark . i. Cyclic codes are the ideals of the ring R_n .

- ii. Every ideal in R_n is a *principal ideal*.
- ie. R_n is a PID.

Theorem 6.1.3. a linear code C in $\mathbb{F}_q^n$ is cyclic iff C is an ideal in $\mathbb{F}_q[x]/(x^n-1)$.

Proof. If C ideal in $\mathbb{F}[x]/(x^n-1)$ and $c(x) = c_0 + c_1x + \dots + c_{n-1}x^{n-1} \in C$, then

$$\begin{aligned} x \cdot c(x) &= c_0x + c_1x \cdot x + \dots + \underbrace{c_{n-1}x^{n-1} \cdot x}_{=c_{n-1} \cdot 1 \pmod{x^n-1}} \\ &= c_{n-1} + c_0x + c_1x^2 + \dots + c_{n-2}x^{n-1} \in C \\ \implies (c_{n-1}, c_0, c_1, \dots, c_{n-2}) &\in C \end{aligned}$$

Conversely, if C cyclic, then $\forall c(x) \in C$, $x \cdot c(x) \in C$.

Therefore $x^i \cdot c(x) \in C \forall i \in [n]$

and since C is linear, $a(x) \cdot c(x) \in C$

ie.

$$a_0 \cdot c(x) + a_1 \cdot x \cdot c(x) + a_2 \cdot x^2 \cdot c(x) + \dots + a_{n-1} \cdot x^{n-1} \cdot c(x) \in C$$

$$\forall a(x) \in \mathbb{F}_q[x]/(x^n-1). \quad \square$$

Corollary . $\mathbb{F}_q[x]/(x^n-1)$ is a PID (Principal Ideal Domain).

Thus C consists of the multiples of $g(x) \in C$ monic and of lowest degree.

Therefore,

$$\implies g(x) \text{ is the } \textit{generator polynomial} \text{ (more at 12.9) of } C: C = \langle g(x) \rangle.$$

$$\implies g(x) \text{ is divisor of } x^n - 1, \text{ ie. } g(x) | (x^n - 1).$$

since otherwise $\gcd(x^n - 1, g(x))$ would be a polynomial in C with degree $< \deg(g(x))$, a contradiction.

Let $f(x) \in R_n$, $\langle f(x) \rangle$ the subset of R_n consisting of all multiples of $f(x)$ (reduced modulo $x^n - 1$). ie.

$$\langle f(x) \rangle = \{r(x) \cdot f(x) \mid r(x) \in R_n\}$$

Theorem 12.7. for any $f(x) \in R_n$, the set $\langle f(x) \rangle$ is a cyclic code; named the *code generated by $f(x)$* .

Proof. check conditions (i), (ii) of Theorem 12.6:

- i. if $a(x)f(x), b(x)f(x) \in \langle f(x) \rangle$ then

$$a(x)f(x) + b(x)f(x) = (a(x) + b(x))f(x) \in \langle f(x) \rangle$$

- ii. if $a(x)f(x) \in \langle f(x) \rangle$ and $r(x) \in R_n$, then

$$r(x)(a(x)f(x)) = (r(x) \cdot a(x))f(x) \in \langle f(x) \rangle$$

□

Example 12.8. $C = \langle 1 + x^2 \rangle \in R_3$, with $GF(2)$.

R_3 has $2^3 = 8$ elements.

Multiply $1 + x^2$ by each of the 8 elements of R_3 (reducing modulo $x^3 - 1$), obtaining 4 distinct codewords:

$$0, 1 + x, 1 + x^2, x + x^2$$

Thus C is the code $\{000, 110, 101, 011\}$.

7.3 Generator polynomial and check polynomial

Theorem 12.9. let C a nonzero cyclic coded in R_n . Then

- i. $\exists!$ $g(x)$ monic, of smallest degree in C
- ii. $C = \langle g(x) \rangle$
- iii. $g(x)$ is a factor of $x^n - 1$

Proof. i. $\exists!$ $g(x)$ monic, of smallest degree in C :

let $g(x), h(x) \in C$ monic and of smallest degree. Then $g(x) - h(x) \in C$ with smaller degree, which is a contradiction if $g(x) \neq h(x)$

- ii. $C = \langle g(x) \rangle$:

let $a(x) \in C$. By division algorithm for $F_q[x]$,

$$a(x) = q(x)g(x) + r(x)$$

with $\deg(r(x)) < \deg(g(x))$.

But $r(x) = a(x) - q(x)g(x) \in C$ by the properties of cyclic codes (Theorem 12.6, $a, g \in C$ and C an ideal)

By minimality of $\deg(g(x))$, must have $r(x) = 0$, therefore

$$\begin{aligned} a(x) &\in \langle g(x) \rangle \forall a(x) \in C \\ \implies C &= \langle g(x) \rangle \end{aligned}$$

- iii. $g(x)$ is a factor of $x^n - 1$:
by division algorithm,

$$x^n - 1 = q(x)g(x) + r(x)$$

with $\deg(r(x)) < \deg(g(x))$.

But then $r(x) = -q(x)g(x) \bmod x^n - 1$, ie. $r(x)$ is a multiple of $g(x)$, and so $r(x) \in \langle g(x) \rangle$.

By the minimality of $\deg(g(x))$, we must have $r(x) = 0$, which implies

$$x^n - 1 = q(x) \cdot g(x) + 0$$

$\implies g(x)$ is a factor of $(x^n - 1)$.

□

Definition (Generator polynomial). let C cyclic code, monic polynomial of least ddegree (given by Theorem 12.9) is called the *generator polynomial* of C .

Remark C may contain other polynomials than the generator polynomial which also generate C .

Example 12.10. All cyclic codes of length 3:

ie. ideals in $\mathbb{F}_2[x]/(x^3 - 1) \implies$ factor $x^3 - 1$ in $\mathbb{F}_2$, every cyclic code corresponds to a divisor of $x^3 - 1$.

$$x^3 - 1 = (x + 1) \cdot (x^2 + x + 1)$$

so by Theorem 12.9(ii):

generator poly	code in R_3	Corresponding code in $V(3, 2)$
1	all of R_3	all of $V(3, 2)$
$x + 1$	$\{0, 1 + x, x + x^2, 1 + x^2\}$	$\{000, 110, 011, 101\}$
$x^2 + x + 1$	$\{0, 1 + x + x^2\}$	$\{000, 111\}$
$x^3 - 1$	$\{0\}$	$\{000\}$

Notice that the 2nd column contains the multiples $(\bmod x^3 - 1)$ of the polynomials at the 1st column.

Lemma 12.11. let $g(x) = g_0 + g_1x + \dots + g_rx^r$ be the generator polynomial. Then g_0 is nonzero.

Proof. by contradiction, let $g_0 = 0$. Then $x^{n-1} \cdot g(x) = x^{-1} \cdot g(x) \in C$ with degree $r - 1$.

This contradicts minimality of $\deg(g(x))$; thus $g_0 \neq 0$.

□

Theorem 12.12. let C cyclic code with generator polynomial $g(x) = g_0 + g_1x + \dots + g_rx^r$. Then $\dim(C) = n - r$, and a GM for C is

$$G = \begin{pmatrix} g_0 & g_1 & g_2 & \cdots & g_r & 0 & 0 & \cdots & 0 \\ 0 & g_0 & g_1 & g_2 & \cdots & g_r & 0 & \cdots & 0 \\ 0 & 0 & g_0 & g_1 & g_2 & \cdots & g_r & \cdots & 0 \\ \vdots & & & & & & & \ddots & \\ 0 & 0 & \cdots & 0 & g_0 & g_1 & g_2 & \cdots & g_r \end{pmatrix}$$

Proof. the $n-r$ rows of G are linear independent because the echelon of nonzero g_0 's with 0's below.

These $n-r$ represent the codewords $g(x), x \cdot g(x), x^2 g(x), \dots, x^{n-r-1} g(x)$.

It remains to show that every codeword in C can be expressed as a linear combination of them.

From Theorem 12.9.(ii), if $a(x) \in C$ then $a(x) = q(x) \cdot g(x)$ for $q(x) \in \mathbb{F}_q[x]$.

Since $\deg(a(x)) < n$, $\deg(q(x)) < n-r$.

Hence

$$\begin{aligned} q(x)g(x) &= (q_0 + q_1x + \dots + q_{n-r-1}x^{n-r-q}) \cdot g(x) \\ &= q_0g(x) + q_1xg(x) + \dots + q_{n-r-1}x^{n-r-q}g(x) \end{aligned}$$

which is the desired linear combination. $\square$

Definition (check polynomial). For generator polynomial $g(x)$, which we know is a factor of $x^n - 1$.

Then, the *check polynomial* $h(x)$ is such that

$$x^n - 1 = g(x)h(x)$$

Since $g(x)$ is monic, so also is $h(x)$. By Theorem 12.12,

$$\deg(g(x)) = n - k \implies \deg(h(x)) = k$$

Theorem 12.14. $c(x) \in R_n$ is $c(x) \in C$ iff $c(x) \cdot h(x) = 0$.

Proof. in R_n : $g(x) \cdot h(x) = x^n - 1 = 0 \pmod{x^n - 1}$,

hence

$$c(x) \in C \xrightarrow{\text{implies}} c(x) = a(x) \cdot g(x)$$

for some $a(x) \in R_n$.

$$\begin{aligned} \implies c(x) \cdot h(x) &= \overbrace{a(x) \cdot g(x)}^{c(x)} \cdot h(x) \\ &\text{(note that } g(x) \cdot h(x) = 0\text{)} \\ &= a(x) \cdot 0 = 0 \end{aligned}$$

$\implies c(x) \cdot h(x) = 0$ as stated.

Conversely, suppose $c(x) \cdot h(x) = 0$.

By division algorithm, $c(x) = q(x) \cdot g(x) + r(x)$ with $\deg(r) < n - k$.

Then, $c(x) \cdot h(x) = 0$ implies

$$\begin{aligned} (q(x) \cdot g(x) + r(x)) \cdot h(x) &= 0 \\ q(x) \cdot g(x) \cdot h(x) + r(x) \cdot h(x) &= 0 \\ \text{recall } g(x) \cdot h(x) &= x^n - 1, \text{ thus,} \\ 0 + r(x) \cdot h(x) &= 0 \\ r(x) \cdot h(x) &= 0 \\ \text{ie. } r(x) \cdot h(x) &= 0 \pmod{x^n - 1} \end{aligned}$$

But $\deg(r(x) \cdot h(x)) < n - k + k = n$, so $r(x) \cdot h(x) = 0$ in $\mathbb{F}_q$.
Hence $r(x) = 0$, thus

$$\begin{aligned} c(x) &= q(x) \cdot g(x) \in C \\ \implies c(x) &\in C \end{aligned}$$

since C is an ideal of R_n and $g(x) \in C$ and $q(x) \in R_n$. □

Theorem 12.15. C cyclic $[n, k]$ -code with check polynomial

$$h(x) = h_0 + h_1x + \dots + h_kx^k$$

Then

i. a PCM for C is

$$H = \begin{bmatrix} h_k & h_{k-1} & h_{k-2} & \cdots & h_0 & 0 & 0 & \cdots & 0 \\ 0 & h_k & h_{k-1} & h_{k-2} & \cdots & h_0 & 0 & \cdots & 0 \\ 0 & 0 & h_k & h_{k-1} & h_{k-2} & \cdots & h_0 & \cdots & 0 \\ & \vdots & & & & & & \vdots & \\ 0 & 0 & \cdots & 0 & h_k & h_{k-1} & h_{k-2} & \cdots & h_0 \end{bmatrix}$$

ii. $C^\perp$ is a cyclic code generated by

$$\bar{h}(x) = h_k + h_{k-1}x + \dots + h_0x^k$$

7.4 Maximal and minimal cyclic codes

Let $x^n - 1 = f_1(x) \cdot f_2(x) \cdot \dots \cdot f_t(x)$, with $f_i(x)$ irreducible polynomials.

Definition 6.1.6 (maximal and minimal cyclic codes). the cyclic code generated by $f_i(x)$ is called a *maximal cyclic code* (since it's a maximal ideal), denoted M_i^+ .

The code generated by $(x^n - 1)/f_i(x)$ is called a *minimal cyclic code*, denoted M_i^- . Also named *irreducible code*.

ie.

$$M_i^+ = \langle f_i(x) \rangle, \quad M_i^- = \left\langle \frac{(x^n - 1)}{f_i(x)} \right\rangle$$

Definition (Zeros of a cyclic code). Let $x^n - 1 = f_1(x) \cdot \dots \cdot f_t(x)$. Let β_i be a zero of $f_i(x)$ in some extension field $E \supset \mathbb{F}_q$,

ie. $f_i(\beta_i) = 0$, with $f_i(x)$ the minimal polynomial of β_i .

$\implies M_i^+$ is the set of polynomials $c(x)$ for which $c(\beta_i) = 0$.

Specify a cyclic code by requiring all codewords to have certain zeros, ie. let the set $\alpha_1, \alpha_2, \dots, \alpha_s$; define the code C by

$$c(x) \in C \text{ iff } c(\alpha_i) = 0 \ \forall i \in [s]$$

then C cyclic, and the generator polynomial is the lcm of the minimal polynomials of $\alpha_1, \alpha_2, \dots, \alpha_s$.

Take the $sm \times n$ matrix with columns $1, \alpha_i, \alpha_i^2, \dots, \alpha_i^{n-1}$, denoted H , with entries in $\mathbb{F}_q$. Then

$$\begin{aligned} cH^T &= 0, \text{ with } c = (c_0, c_1, \dots, c_{n-1}) \\ c(\alpha_i) &= 0 \ \forall i \in [s], \text{ with } c(x) \in C \end{aligned}$$

7.5 Hamming cyclic codes

Theorem 6.3.1. let $n := (q^m - 1)/(q - 1)$, let β a primitive n -th root of unity in $\mathbb{F}_{q^m}$. Let $(m, q - 1) = 1$.

The cyclic code $C := \{c(x) | c(\beta) = 0\}$ is equivalent to the $[n, n - m]$ Hamming code over $\mathbb{F}_q$.

Proof. since $n = (q - 1) \cdot (q^{m-2} + 2q^{m-3} + \dots + m - 1) + m$ have $(n, q - 1) = (m, q - 1) = 1$.

Therefore $\beta^{i(q-1)} \neq 1 \ \forall i \in [n]$, ie. $\beta^i \neq \mathbb{F}_q \ \forall i \in [n]$.

Columns of H , which are the representations of $1, \beta, \beta^2, \dots, \beta^{n-1}$ as vectors in $\mathbb{F}_{q^m}$, are pairwise linearly independent over $\mathbb{F}_q$.

So H is the PCM of an $[n, n - m]$ Hamming code. $\square$

Example 6.3.2. smallest extension of $\mathbb{F}_2$ which contains a primitive 9-th root of unity is $\mathbb{F}_{2^6}$.

Let α a primitive element of the field, then $\alpha^6 3 = 1$, and $\beta := \alpha^7$ is a primitive 9-th root of unity.

Then, the minimal polynomial of β has the zeros

$$\beta, \beta^2, \beta^4, \beta^8, \beta^{16} = \beta^7, \beta^{14} = \beta^5$$

$$\implies \text{min. polynomial: } (x^9 - 1)/(x^3 - 1) = x^6 + x^3 + 1.$$

$$\text{So } (x^9 - 1) = \underbrace{(x - 1)}_{f_1(x)} \underbrace{(x^2 + x + 1)}_{f_2(x)} \underbrace{(x^6 + x^3 + 1)}_{f_3(x)}.$$

The code M_3^+ has pairwise independent columns in H , ie. min.distance ≥ 3 .

M_3^+ consists of codewords $(c_0 c_1 c_2 \ c_0 c_1 c_2 \ c_0 c_1 c_2)$, so $d = 3$.

The code M_3^- has check polynomial $x^6 + x^3 + 1$, so it's a $[9, 6]$ -code.

Since $x^3 - 1 \in M_3^-$ (is a codeword), the distance is 2.

7.6 Idempotent of a cyclic code

Theorem 6.4.1. $\exists! c(x) \in C$ which is an identity element for C .

Proof. let $g(x)$ generator polynomial, $h(x)$ check polynomial.

ie. $g(x) \cdot h(x) = x^n - 1$ in $\mathbb{F}_q[x]$.

Since $x^n - 1$ has no multiple zeros (ie. has simple zeros),

we have $(g(x), h(x)) = 1$, hence by the Bezout identity:

$$\exists a(x), b(x) \in \mathbb{F}_q[x] \text{ s.th. } a(x)g(x) + b(x)h(x) = \gcd(g, h) = 1$$

Define $c(x) := a(x)g(x) = 1 - b(x)h(x)$.

Clearly $c(x) \in C$. Furthermore, if $p(x)g(x) \in C$ (ie. $p(x) \cdot g(x)$ is any codeword in C), then

$$\begin{aligned} c(x)p(x)g(x) &= \underbrace{(1 - b(x)h(x))}_{c(x)} p(x)g(x) \\ &= p(x)g(x) - b(x)h(x)p(x)g(x) \\ (\text{recall : } g(x)h(x) &= x^n - 1) \\ &= p(x)g(x) - b(x)p(x) \underbrace{g(x)h(x)}_{=x^n-1} \\ &= p(x)g(x) \pmod{x^n - 1} \end{aligned}$$

So $c(x)$ is an identity element for C , and hence is unique. $\square$

Recall:

nilpotent $a \in A$ s.th. $a^n = 0$ for $n > 0$

idempotent $e \in A$ s.th. $e^2 = e$

Definition 6.4.2 (primitive idempotent). the idempotent of an irreducible cyclic code M_i^- is called a *primitive idempotent*, denoted by $\theta_i(x)$.

Theorem 6.4.3. if C_1, C_2 are cyclic codes with idempotents $c_1(x), c_2(x)$, then

- i. $C_1 \cap C_2$ has idempotent $c_1(x) \cdot c_2(x)$
- ii. $C_1 + C_2$, ie. the set of all words $a + b$ with $a \in C_1$, $b \in C_2$, has idempotent $c_1(x) + c_2(x) + c_1(x) \cdot c_2(x) \in C_1 + C_2$.

Theorem 6.4.4. for the primitive idempotent we have:

- i. $\theta_i(x)\theta_j(x) = 0$ if $i \neq j$
- ii. $\sum_{i=1}^t \theta_i(x) = 1$
- iii. $1 + \theta_{i_1}(x) + \theta_{i_2}(x) + \dots + \theta_{i_r}(x)$ is the idempotent of the code with generator $f_{i_1}(x) \cdot f_{i_2}(x) \cdot \dots \cdot f_{i_r}(x)$.

7.7 Trace representation

Definition 1.1.29. $q = p^r$, trace: $Tr : \mathbb{F}_q \longrightarrow \mathbb{F}_p$.

If $\zeta \in \mathbb{F}_q$, then

$$Tr(\zeta) := \zeta + \zeta^p + \zeta^{p^2} + \dots + \zeta^{p^{r-1}} \in \mathbb{F}_p$$

Theorem 1.1.30. properties of the trace function:

- i. $\forall \zeta \in \mathbb{F}_q, Tr(\zeta) \in \mathbb{F}_p$
- ii. there are elements $\theta \in \mathbb{F}_p$ such that $Tr(\zeta) \neq 0$
- iii. Tr is a linear mapping.

Proof. i. by definition $(Tr(\zeta))^p = Tr(\zeta)$

ii. the equation $x + x^p + \dots + x^{p^{r-1}} = 0$ can not have q roots in $\mathbb{F}_q$

iii. since $(\zeta + \eta)^p = \zeta^p + \eta^p$ and $\forall a \in \mathbb{F}_p$ we have $a^p = a$

□

Corollary . Tr takes every value $p^{-1}q$ times, and the polynomial $x + x^p + \dots + x^{p^{r-1}}$ is a product of minimal polynomials.

Theorem 6.5.1. let k be the multiplicative order of $p \bmod n$, $q = p^k$, β a primitive n -th root of unity in $\mathbb{F}_q$.

Then the set

$$V := \{c(\zeta) := (Tr(\zeta), Tr(\zeta\beta), \dots, Tr(\zeta\beta^{n-1})) \mid \zeta \in \mathbb{F}_q\}$$

is an $[n, k]$ irreducible code over $\mathbb{F}_p$.

Proof. by Theorem 1.1.30, V is a linear code, and $c(\zeta\beta^{-1})$ is a cyclic shift of $c(\zeta)$; hence V is a cyclic code.

Since β is in no subfield of $\mathbb{F}_q$, we know that β is a zero of an irreducible polynomial $h(x) = h_0 + h_1x + \dots + h_kx^k$.

If $c(\zeta) = (c_0, c_1, \dots, c_{n-1})$, then

$$\sum_{i=0}^k c_i h_i = Tr(\zeta h(\beta)) = Tr(0) = 0$$

since β is a zero of $h(x)$, so that $h(\beta) = 0$.

So we observe $c(x)h(x) = 0$, from Theorem 12.14 this means that $h(x)$ is the check polynomial,

ie. we have a parity check equation for the code V .

Since $h(x)$ irreducible, $x^k \cdot h(x^{-1})$ is the check polynomial for V , therefore, V is an irreducible cyclic $[n, k]$ -code. □

7.8 Mattson-Solomon Polynomial

Definition (Mattson-Solomon polynomial). Let β primitive n -th root of unity in the extension field E of $\mathbb{F}_q$.

Let T the set of polynomials over E with degree at most $n - 1$, ie.

$$T = \{p(x) \in E \mid \deg(p(x)) \leq n - 1\}$$

Define $\Phi : T \longrightarrow T$ as:

let $a(x) \in T$, then $A(X) = (\Phi_a)(X)$ is defined by

$$A(X) := \sum_{j=1}^n a(\beta^j) X^{n-j}$$

If $a = (a_0, a_1, \dots, a_{n-1})$, the polynomial $A(X)$ obtained from $a_0 + a_1x + \dots + a_{n-1}x^{n-1}$ is called the *Mattson-Solomon polynomial* of the vector a .

Lemma 6.5.3. the inverse of Φ is given by

$$a(x) = n^{-1}(\Phi A)(x^{-1}) \pmod{x^n - 1}$$

Proof.

$$A(\beta^k) = \sum_{j=1}^n \sum_{i=0}^{n-1} a_i \beta^{ij} \beta^{-kj} = \sum_{i=0}^{n-1} a_i \sum_{j=1}^n \beta^{(i-k)j} = na_k$$

□

Remark . Notation:

$\circ$: multiplication of polynomials $\text{mod}(x^n - 1)$

$*$: defined by

$$\left(\sum a_i x^i\right) * \left(\sum b_i x^i\right) := \sum a_i b_i x^i$$

Φ is an isomorphism of the ring $(T, +, \circ)$ onto the ring $(T, +, *)$.

Lemma 6.5.4. let V a cyclic code over $\mathbb{F}_q$ generated by

$$g(x) = \prod_{k \in K} (x - \beta^k)$$

Suppose $\{1, 2, \dots, d-1\} \subset K$ and $a \in V$.

Then the degree of the Mattson-Solomon polynomial A of a is at most $n - d$.

Proof. $a(\beta^j) = 0$ for $j \in [d]$ since $g(x) \mid a(x)$.

Thus at the equation $A(X) := \sum_{j=1}^n a(\beta^j) X^{n-j}$, it will be 0 between $1 \leq j \leq d-1$, so that degree is $n - d$. □

Theorem 6.5.5. if there are r n -th roots of unity which are zeros of the Mattson-Solomon polynomial A of a word a , then $w(a) = n - r$.

Proof. consequence of Lemma 6.5.3. □

8 BCH Codes

Definition 6.6.1 (BCH Code). a cyclic code of length n over $\mathbb{F}_q$ is called *BCH code of designed distance δ* if its generator $g(x)$ is the lcm of the minimal polynomials of $\beta^l, \beta^{l+1}, \dots, \beta^{l+\delta-2}$ for some l , where β is a primitive n -th root of unity.

Usually $l = 1$; if $n = q^m - 1$ (ie. β a primitive element of $\mathbb{F}_q^m$), then the BCH code is called *primitive*.

Theorem 6.6.2 (BCH Bound). the minimum distance of a BCH code with designed distance d is $\geq d$.

Proof. wlog. take $l = 1$. By Lemma 6.5.4, the degree of the Mattson-Solomon polynomial A of a codeword c is $\deg(A) \leq n - d$.

Therefore, in theorem 6.5.5 we have $r \leq n - d$, ie. $w(c) \geq d$.

(recall: r is the amount of n -th roots of unity which are zeros of the M-S polynomial A) $\square$

Definition (Defining set). If $A = \{\alpha^{i_1}, \dots, \alpha^{i_l}\}$ is a set of n -th roots of unity such that

$$c(x) \in C \iff \forall \zeta \in A \ c(\zeta) = 0$$

we say that A is the *defining set* for C .

If A is the *maximal* defining set for C , we call A *complete*.

Definition 6.6.3. $M(A)$ or $M(\alpha^{i_1}, \dots, \alpha^{i_l})$: matrix $l \times n$ that has $1, \alpha^{i_k}, \alpha^{2i_k}, \dots, \alpha^{(n-1)i_k}$ as its k -th rows, ie.

$$M(A) = \begin{bmatrix} 1 & \alpha^{i_1} & \alpha^{2i_1} & \dots & \alpha^{(n-1)i_1} \\ 1 & \alpha^{i_2} & \alpha^{2i_2} & \dots & \alpha^{(n-1)i_2} \\ \vdots & & & & \vdots \\ 1 & \alpha^{i_l} & \alpha^{2i_l} & \dots & \alpha^{(n-1)i_l} \end{bmatrix}$$

$M(A)$ is the PCM corresponding to A .

Theorem 6.6.13. The primitive binary BCH code C of length $n = 2^m - 1$ and designed distance $\delta = 2^l - 1$ has minimum distance δ .

Corollary 6.6.14. a primitive BCH code of designed distance δ has distance $d \leq 2\delta - 1$.

Corollary 6.6.16. the minimum weight of a primitive binary BCH code is odd.

9 Reed-Solomon Codes

Definition 6.8.1 (Reed-Solomon code). a *Reed-Solomon code* (RS code) is a primitive BCH code of length $n = q - 1$ over $\mathbb{F}_q$.

With generator

$$g(x) = \prod_{i=1}^{d-1} (x - \alpha^i)$$

where α is primitive in $\mathbb{F}_q$.

Remark . observe:

- by 6.6.2, minimum distance $\geq d$
- by 6.2, has dimension $k = n - d + 1$

Therefore, Corollary 5.2.2 implies that minimum distance is d and the RS code is a MDS.

This is useful for when errors are in bursts (ie. several errors together).

Let $n = q - 1$, α a primitive element of $\mathbb{F}_q$.

$$a = (a_0, a_1, \dots, a_{k-1}) \in \mathbb{F}_q^k \iff a(x) = a_0 + a_1x + \dots + a_{k-1}x^{k-1}$$

Then

$$C = \{(c_0, c_1, \dots, c_{n-1}) \mid c_i = a(\alpha^i), 0 \leq i < n, a \in \mathbb{F}_q^k\}$$

is the RS code with $d = n - k + 1$.

To see it: C is cyclic. By definition of C and Lemma 6.5.3, implies that a codeword c has $n \cdot a(x)$ as its Mattson-Solomon polynomial.

Since $\deg(a(x)) \leq k - 1$,

$$c(\alpha^i) = 0 \quad \forall i \in [1, n - k]$$

Extend the words of C by adjoining a symbol $c_n = a(0)$, then

$$\sum_{i=0}^n c_i = a_0q = 0$$

$\implies$ we obtain $\overline{C}$.

If $c_n = 0$, ie. $c(1) = 0$, then the word has weight $\geq n - k + 2$. Also true if $c_n \neq 0$. $\implies$ so $\overline{C}$ is also a MDS code.

Definition 6.8.2. the generalized R-S code, $GRS_k(a, v)$, has codewords all

$$(v_1f(\alpha_1), v_2f(\alpha_2), \dots, v_nf(\alpha_n))$$

where f runs through the set of polynomials of degree $< k$ in $\mathbb{F}_{q^m}[X]$, ie. $f \in \mathbb{F}_{q^m}[X]^{<k}$.

A GRS and its dual are MDS codes.

Theorem 6.8.3. the dual of an MDS code is also an MDS code.

Proof. let $G = (I_k \ P)$ be GM of C .

Since C has min. weight d , every set of $d - 1 = n - k$ columns of the PCM $H := (-P^T \ I_{n-k})$ is linearly independent.

Hence every square submatrix of H is nonsingular, ie. no codeword of $C^\perp$ has $n - k$ zeros.

So $C^\perp$ is an $[n, n - k, k + 1]$ code, ie. MDS. $\square$

9.1 Example of Hamming code (cyclic) and of Reed-Solomon code (cyclic and MDS)

(7,4)-Hamming code over $\mathbb{F}_2$

$$(n, k) \implies n = 7(\text{length}), k = 4(\text{message length})$$

Generator polynomial: $g(x) = x^3 + x + 1$.

Parity bits: $r = 3$

Message: $m(x) \in \mathbb{F}_2[X]^{\leq 3}$, encoded as $c(x) = m(x)g(x) \pmod{(x^7 - 1)}$

eg: for $m(x) = x^3 + x + 1$,

$$\begin{aligned} c(x) &= (x^3 + x + 1) \cdot (x^3 + x + 1) \pmod{(x^7 - 1)} = x^6 + x^2 + 1 \\ &\implies c = (1000101) \end{aligned}$$

Hamming code is cyclic:

when $c(x) \in C$, then $x \cdot c(x) \pmod{(x^n - 1)} \in C$,

because $C = \langle g(x) \rangle$, ie. all codewords are multiples of $g(x)$,

so for $c(x) = m(x) \cdot g(x)$, then

$$x \cdot c(x) = x \cdot m(x) \cdot g(x)$$

is still a multiple of $g(x)$, hence a valid codeword.

(6,4)-Reed-Solomon code over $\mathbb{F}_7$

$$n = 6, k = 4, \text{ minimum distance: } d = n - k + 1 = 6 - 4 + 1 = 3$$

Choose a primitive element, $\alpha = 3 \in \mathbb{F}_7$.

Generator polynomial:

$$\begin{aligned} g(x) &= (x - \alpha) \cdot (x - \alpha^2) \\ &= x^2 - 2x - 3x + 6 \\ &= x^2 - 5x + 6 = x^2 + 2x + 6 \pmod{7} \end{aligned}$$

message: $m(x) = 1 + 2x + x^2 + x^3$

Then,

cyclic form: $c(x) = m(x)g(x) \pmod{(x^6 - 1)}$
 $c(x) \in C$ iff $c(\alpha^i) = 0 \ \forall i \in [n - k]$

evaluation form: $c = (m(1), m(\alpha), m(\alpha^2), \dots, m(\alpha^6))$

RS code is cyclic. And it is MDS:

- a polynomial of degree $< k$ is uniquely determined by k points
- if two polynomials agree in k positions, are the same polynomial
 $\implies$ maximum number of equal positions between distinct codewords = $k - 1$

Therefore, $d = n - (k - 1) = n - k + 1$, ie. Singleton bound (with equality!).

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